POWER CONTROL

prepared by:

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

100%

LABORATORY

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

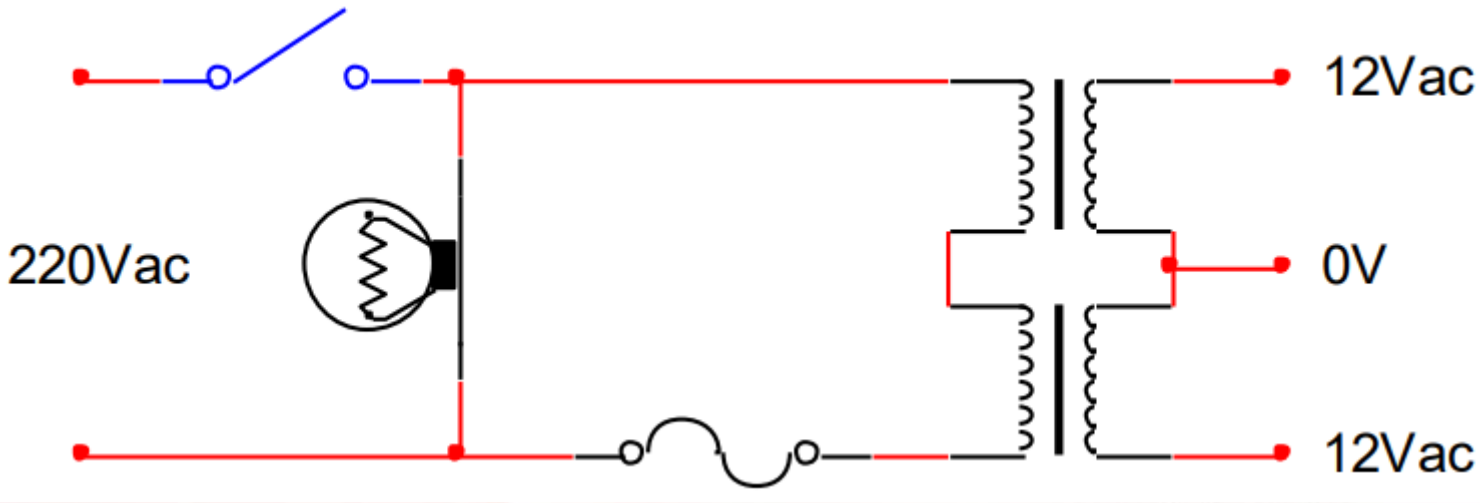
Mounting Power Control Components

- Schematic and Wiring Diagram
- Parts List

Tools and Instruments

MOUNTING POWER CONTROL COMPONENTS

SCHEMATIC DIAGRAM

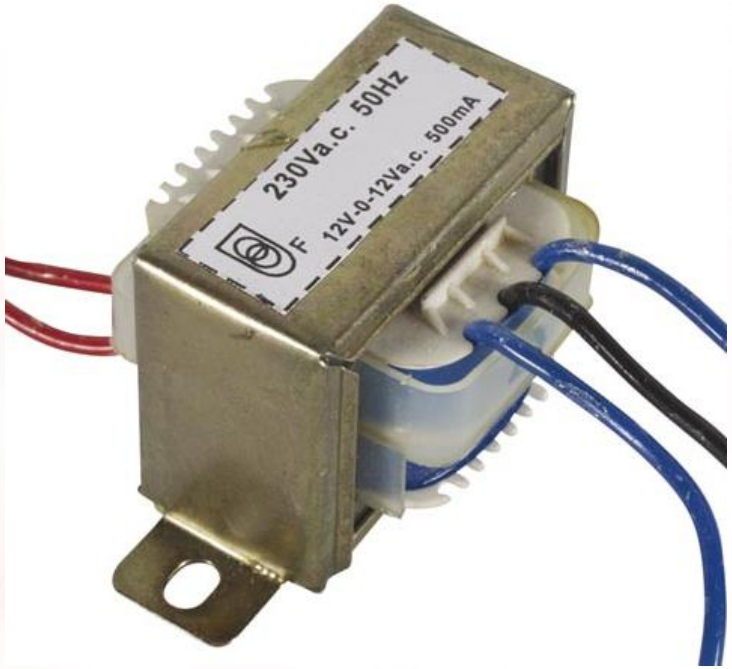


switch

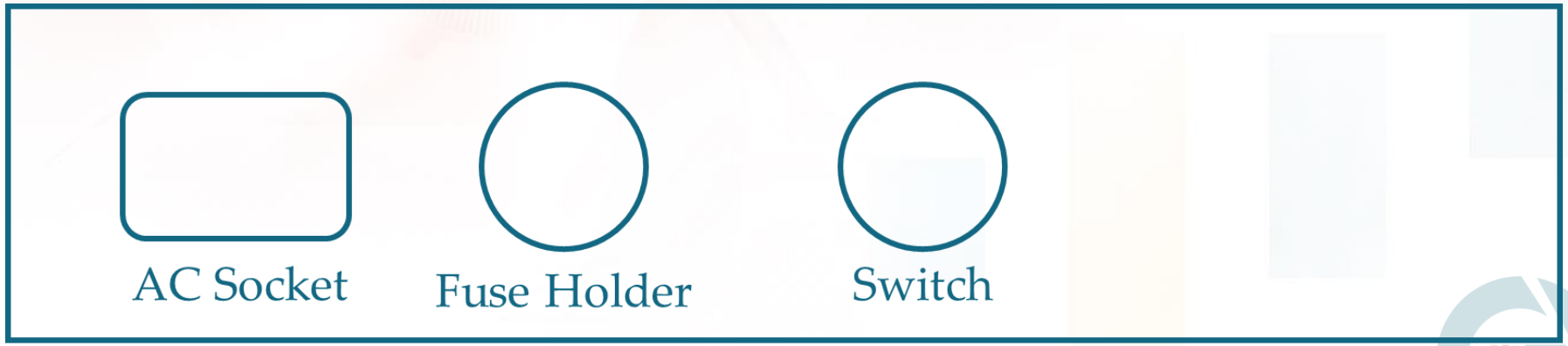
AC Power Cord Socket

Fuse Holder

WIRING DIAGRAM



FRONT VIEW



PARTS LIST

Quantity	Description
1pc	Case
1pc	12-0-12V, 500mA Transformer
1pc	AC switch w/ lamp
1pc	AC chord + Socket
1pc	Fuse Holder + 3A Fuse
1m	Shrinkable Tube (for AC switch/fuse)
1m	Soldering Lead
1m	Solid Wire #20AWG

PARTS LIST

Quantity	Description
2pcs	Diode, 1N4001
1pc	Electrolytic Capacitor 100uF/16V
3pcs	Resistor 1K, 1/4W
1pc	Copper Board 3”x 4”
1bot	Ferric Chloride
1m	Soldering Lead

TOOLS AND INSTRUMENTS

TOOLBOX

Portable container or **storage** unit designed to hold and organize tools and other items used for various tasks and repairs

SOLDERING IRON

20W – 30W

Supplies heat to **melt solder** so that it can flow into the joint between two workpieces

DESOLDERING PUMP

Also known as **solder sucker**

Sucks the **liquid/molten solder** from the joint where the components are mounted

NEEDLE-NOSE PLIERS

Long and narrow, with **pointy**, curved or angled tips that have a strong grip

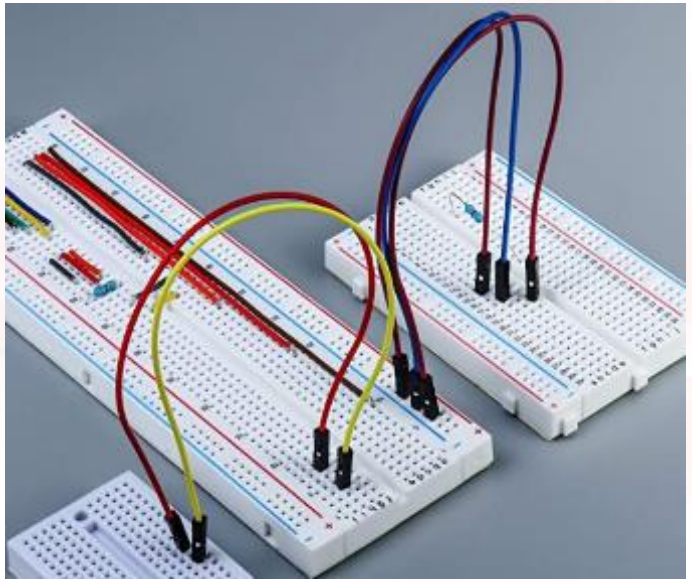
DIAGONAL WIRE CUTTERS AND END CUTTERS (NIPPERS)

Jaws intersects the joint rivet at an angle or "on a **diagonal**"

PRECISION SCREWDRIVER SET

Set of **screwdrivers**

BREADBOARDS



A thin plastic board used to **hold** electronic components (transistors, resistors, chips, etc.) that are wired together

JUMPER WIRES

Connect **two points** to each other without soldering

MINI DRILL

Making round **holes** or driving fasteners

MULTIMETER

Also known as a **volt-ohm meter**

Handheld tester used to measure electrical **voltage**, **current**, **resistance**, and other values

TWEEZERS

Picking up objects too **small** to be easily handled with the human fingers

WIRE STRIPPER

Remove the **insulation** from electric wires

OSCILLOSCOPE

Display and analyze the **waveform** of electronic signals

POWER SUPPLY

Powers the circuit to test individual components electrical responses

LABORATORY